Mathematical Problems in Deep Learning: Homeworks #2 Solutions

1. **Randomized Response, re-revisited:** We will see some generalizations of randomized response, beyond just binary alphabets. I will informally and vaguely describe an algorithm, you must rigorously define and specify the algorithm and prove that it is ϵ-differentially private.

   (a) Assume $X_i \in \{1, \ldots, k\}$ for the remainder of this problem. The vector $(Y_1, \ldots, Y_n) \in \{1, \ldots, k\}^n$ is output, where Y_i is equal to X_i with probability proportional to $g(\epsilon)$ (for some function g which you must specify), and equal to each $s \in \{1, ..., k\} \setminus X_i$ with probability proportional to 1. Specify the algorithm rigorously (define probability of randomized response) and prove that it is ϵ-DP.

   (b) Here is another way to generalize randomized response. The vector $(Y_1, \ldots, Y_n) \in \{0, 1\}^{kn}$ is output. $Y_i \in \{0, 1\}^k$ is a vector generated in the following manner: each X_i is first converted to a "one-hot" vector $\in \{0, 1\}^k$, where coordinate j is 1 if $j = X_i$ and 0 otherwise. Y_i generated from X_i by applying a bitwise randomized response (with appropriate parameter). Specify the algorithm rigorously (define probability of randomized response) and prove that it is ϵ-DP.

   Solution:

   (a) We can set $g(\epsilon) = e^\epsilon$, so that

   $$\begin{aligned} Pr(Y_i = X_i) =& \frac{e^\epsilon}{e^\epsilon + (k-1)} \\ Pr(Y_i = s) =& \frac{1}{e^\epsilon + (k-1)} \end{aligned}$$

   for $s \neq X_i$ The probability ratio is always bounded by e^ϵ.

   (b) Without loss of generality, consider X and X' where $X_1 = (1, 0, \ldots, 0)$ and $X_1' = (0, 1, 0, \ldots, 0)$ (and all others are the same $X_i = X_i'$).

   $$\frac{Pr(Y_1 = (b_1, \ldots, b_k))}{Pr(Y_1' = (b_1, \ldots, b_k))} = \frac{Pr(Y_{11}, Y_{12} = b_1, b_2)}{Pr(Y_{11}', Y_{12}' = b_1, b_2)}.$$

   When the bit-flipping probability is $q < 1/2$, the maximum ratio would be $(1-q)^2/q^2$ which should be bounded by ϵ. Thus, the flipping probability should be

   $$q = \frac{e^{\epsilon/2}}{1 + e^{\epsilon/2}}.$$

2. **Approximate DP:** Consider the following mechanisms M that takes a dataset $x \in [0, 1]^n$ and returns an estimate of the mean $\bar{x} = (\sum_{i=1}^n x_i)/n$.

   (M1) $M_1(x) = \bar{x} + [Z]_{-1}^1$, for $Z \sim \text{Lap}(2/n)$.

(M2)

$$M_2(x) = \begin{cases} 1 & \text{w.p. } \overline{x} \\ 0 & \text{w.p. } 1 - \overline{x}. \end{cases}.$$

The above mechanisms do not meet the definition of $(\epsilon, 0)$-differential privacy. For those mechanisms, calculate the smallest value of δ (again possibly as a function of n) for which they satisfy (ϵ, δ) differential privacy for a finite value of ϵ.

Solution.

(a) Suppose $\bar{x} = 1/n$ and $\bar{x'} = 0$. For any a, we have

$$Pr(a \leq 1/n + [Z]_{-1}^{1} \leq 1 + 1/n) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1 + 1/n) + \delta.$$

If $a > 1$, we have

$$Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1) \leq \delta$$

which implies

$$\begin{aligned} \delta =& Pr(1 - 1/n \leq [Z]_{-1}^{1} \leq 1) \\ =& Pr(1 - 1/n \leq Z) \\ =& \frac{1}{2} e^{-(n/2)(1-1/n)} \\ =& \frac{1}{2} e^{-n/2+1/2}. \end{aligned}$$

On the other hand, if $a < 1$, we have

$$\begin{aligned} & Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1 + 1/n) + \delta \\ \Leftrightarrow & Pr(a - 1/n \leq [Z]_{-1}^{1} \leq 1 - 1/n) \leq e^{\epsilon} Pr(a \leq [Z]_{-1}^{1} \leq 1) \end{aligned}$$

which is true for $\epsilon = 1/2$. Thus, the above mechanism is $(1/2, 1/2e^{-n+1/2})$-DP. Note that you should specify ϵ as well since the mechanism is not (ϵ, δ)-DP for $\epsilon < 1/2$.

(b)

$$Pr(M_2(x) = 1) \leq e^{\epsilon} Pr(M_2(x') = 1) + \delta \Leftrightarrow \bar{x} \leq e^{\epsilon} \bar{x'} + \delta.$$

Thus, the δ should be

$$\delta = \max_{x,x'} \bar{x} - e^{\epsilon} \bar{x'} = 1/n.$$

Thus, the above mechanism is $(0, 1/n)$-DP. Note that you should clearly mention that the mechanism is (ϵ, δ)-DP for all ϵ.

3. **Regression:** Consider a dataset where each of its n rows is a pair of real numbers (x_i, y_i), each from an interval $[-b, b]$. Suppose we wish to find a best-fit linear relationship $y_i \approx \beta x_i$ between the y's and the x's. Non-privately, a standard way to estimate β is via the OLS regression formula

$$\hat{\beta} = \hat{\beta}(x, y) = \frac{S_{xy}}{S_{xx}} = \frac{\sum_i x_i y_i}{\sum_i x_i^2}.$$

This is called *ordinary least-squares (OLS)* regression, since $\hat{\beta}$ is the minimizer of the mean-squared residuals

$$\frac{1}{n}\sum_i (y_i - \hat{\beta} x_i)^2.$$

(a) Show that the function $\hat{\beta}(x, y)$ has infinite global sensitivity, and hence we cannot get a useful DP estimate of it via a direct application of the Laplace or Gaussian mechanisms.

(b) Show that S_{xy} and S_{xx} have global sensitivity that is bounded solely as a function of b, and hence each of these can be approximated in a DP manner using the Laplace mechanism.

(c) Using Part 3b with basic composition and post-processing, devise and implement an ϵ-DP algorithm for approximating $\hat{\beta}$ on an arbitrary dataset with $x_i, y_i \in \mathbb{R}$. In addition to the dataset $((x_1, y_1), \ldots, (x_n, y_n))$, your implementation should take as input parameters a clipping bound b and the privacy-loss parameter ϵ.

Solution.

(a) Let $X = \{(1, 0), \ldots, (0, 0)\}$ and $X' = \{(u, v), (0, 0), \ldots, (0, 0)\}$. Then, $\hat{\beta}(X) = 0$, and $\hat{\beta}(X') = v/u$. Since u can be arbitrarily close to 0 and the difference v/u is unbounded.

(b) Sensitivity of S_{xy} is $b \cdot b - b \cdot (-b) = 2b^2$ while sensitivity of S_{xx} is $b \cdot b - 0 \cdot 0 = b^2$.

(c) Based on composition theorem, you can mix two privatization with $\alpha\epsilon$ and $(1-\alpha)\epsilon$. In other words, compute $\tilde{S}_{xx}(X) = S_{xx}(X) + Z_{xx}$ where Z_{xx} be Laplace random variable with parameter $b^2/\alpha\epsilon$. Then, compute $\tilde{S}_{xy}(X) = S_{xy}(X) + Z_{xy}$ where Z_{xy} be a Laplace random variable with parameter $2b^2/(1-\alpha)$. Then, we compute $\tilde{\beta} = \tilde{S}_{xy}/\tilde{S}_{xx}$. The combining $\tilde{S}_{xx}$ and $\tilde{S}_{xy}$ is ϵ-DP due to composition theorem, and the final division does not affect the overall privacy due to post processing property.

Finally, you need to discuss the best α that minimizes the (any reasonable) loss between β and $\hat{\beta}$.

4. For $f : \mathcal{X}^n \to \mathbb{R}$ with ℓ_1 sensitivity Δ, the truncated Laplacian mechanism is

$$\mathcal{M}(X) = f(X) + Z$$

where Z is $\mathrm{LapTr}(\lambda, \tau)$. Recall that the Truncated Laplacian random variable has pdf

$$f_Z(z) = \begin{cases} C\exp(-|z|/\lambda) & \text{if } |z| \le \tau \\ 0 & \text{if } |z| > \tau \end{cases}$$

where C is a normalization constant.

(a) Find C.

(b) Show that the Truncated Laplacian mechanism is (ε, δ)-DP. You may assume $\lambda = \Delta/\varepsilon$ and $\tau < \Delta$, but you need to specify δ.

Solution.

(a) Note that

$$\begin{aligned}\int_{-\tau}^{\tau} \exp(-|z|/\lambda)\,dz &= 2\int_0^{\tau} \exp(-z/\lambda)\,dz \\ &= 2\lambda(1 - \exp(-\tau/\lambda)).\end{aligned}$$

Thus,

$$C = \frac{1}{2\lambda(1 - \exp(-\tau/\lambda))}.$$

(b) Let $\lambda = \Delta/\varepsilon$. For y such that $|f(X) - y| \le \tau$ and $|f(X') - y| \le \tau$, the privacy loss random variable

$$\begin{aligned}I_{X,X'}(y) &= \log \frac{\Pr[\mathcal{M}(X) = y]}{\Pr[\mathcal{M}(X') = y]} \\ &= -\frac{|f(X) - y|}{\lambda} + \frac{|f(X') - y|}{\lambda} \\ &\le \frac{|f(X) - f(X')|}{\lambda} \\ &\le \frac{\Delta}{\lambda} \\ &= \varepsilon.\end{aligned}$$

Now, consider the probability of bad events

$$\begin{aligned}
\Pr[|f(X)-y|>\tau \text{ or } |f(X')-y|>\tau] &= \Pr[|Z|>\tau \text{ or } |f(X')-f(X)-Z|>\tau] \\
&\le \Pr[|\Delta - Z|>\tau] \\
&= \Pr[Z>\tau+\Delta \text{ or } Z<\Delta-\tau] \\
&= \int_{-\tau}^{\Delta-\tau} C\exp(-|z|/\lambda)\,dz \\
&= -C\lambda\left(\exp(-\tau/\lambda)-\exp(-(\tau-\Delta)/\lambda)\right) \\
&= -C\lambda\exp(-\tau/\lambda)(1-\exp(\Delta/\lambda)) \\
&= \frac{-\exp(-\tau/\lambda)(1-\exp(\Delta/\lambda))}{2(1-\exp(-\tau/\lambda))}.
\end{aligned}$$

Thus, it is (ε,δ)-DP when

$$\delta = \frac{\exp(-\tau\varepsilon/\Delta)(\exp(\varepsilon)-1)}{2(1-\exp(-\tau\varepsilon/\Delta))}.$$

5. Let $S = \{(x_1,x_2)\in\mathbb{R}^2 : x_1^2+x_2^2=1\}$. Let $X=(X_1,\dots,X_n)$ where $X_i\in S$ for all $1\le i\le n$.

   (a) Consider a mechanism $M_S : S^n \to S$ that returns $M_S(X)=X_I$ for where $I\sim \text{Unif}\{1,\dots,n\}$. Show that M_S cannot be a $(\varepsilon,0)$-DP.

   (b) Show that M_S is $(0,\delta)$-DP. Specify δ.

   (c) Suppose the utility that you are considering is to approximate the empirical mean μ, where $\mu=\frac{1}{n}\sum_{i=1}^n X_i$. Show that

$$\mathbb{E}[\|M_S(X)-\mu\|^2] = 1-\|\mu\|^2.$$

Solution.

(a) Consider $X=((1,0),(0,1),(0,1),\dots,(0,1))$ and $X'=((0,1),(0,1),(0,1),\dots,(0,1))$. Probability of $\Pr[M_S(X)=(1,0)]=1/n$ but $\Pr[M_S(X')=(1,0)]=0$.

(b) It is $(0,1/n)$-DP.

(c) We have

$$\begin{aligned}
\mathbb{E}[\|M_S(X)-\mu\|^2] &= \sum_{j=1}^n \frac{1}{n}\|X_j-\mu\|^2 \\
&= \sum_{j=1}^n \frac{1}{n}\left(\|X_j\|^2 - 2\langle X_j,\mu\rangle + \|\mu\|^2\right) \\
&= 1-\sum_{j=1}^n \frac{1}{n}2\langle \mu,\mu\rangle + \|\mu\|^2 \\
&= 1-\|\mu\|^2.
\end{aligned}$$

6. Suppose $M : \{0,1\}^n \to \mathcal{Y}$ is an (ε, δ)-differentially private mechanism, and let $A : \mathcal{Y} \to \{0,1\}^n$ be an adversary attempting to reconstruct the sensitive bits in a dataset $X \in \{0,1\}^n$ from the output $M(X)$. More precisely, given $M(X) = Y$, the adversary generates $A(Y) = \hat{X} = (\hat{X}_1, \ldots, \hat{X}_n)$, and wants to maximize the fraction of correct guesses

$$\frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(X_i = \hat{X}_i)$$

where

$$\mathbf{1}(X_i = \hat{X}_i) = \begin{cases} 1 & \text{if } X_i = \hat{X}_i \\ 0 & \text{if } X_i \neq \hat{X}_i. \end{cases}$$

Assume the dataset is a random variable $X = (X_1, \ldots, X_n)$ consisting of n independent draws from a Bernoulli(p) distribution, for a known value of $p < 1/2$.

(a) Let $Z_i = \mathbf{1}(X_i = \hat{X}_i)$. Show that

$$\mathbb{E}[Z_i] = \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1-p)$$

(b) Since M is (ε, δ)-DP, for any neighboring dataset $X \sim X'$, we have

$$\Pr[\hat{X}_i = 1 \mid X] \leq e^{\varepsilon} \Pr[\hat{X}_i = 1 \mid X'] + \delta$$

Then, show that

$$\Pr[\hat{X}_i = 1 \mid X_i = 1] \leq e^{\varepsilon} \Pr[\hat{X}_i = 1 \mid X_i = 0] + \delta$$

(c) Prove that the expected fraction of bits that the adversary successfully reconstructs is not much larger than the trivial bound of $1-p$, which could be achieved by always guessing the all-zero dataset. Specifically, show that:

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} Z_i\right] \leq e^{\varepsilon} \cdot (1-p) + \delta.$$

Solution.

(a) We have

$$\begin{aligned}
\mathbb{E}[Z_i] &= \Pr[Z_i = 1] \\
&= \Pr[\hat{X}_i = X_i] \\
&= \Pr[\hat{X}_i = X_i \mid X_i = 1] \Pr[X_i = 1] + \Pr[\hat{X}_i = X_i \mid X_i = 0] \Pr[X_i = 0] \\
&= \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1-p).
\end{aligned}$$

(b)

$$
\begin{aligned}
&\Pr[\hat{X}_i = 1 \mid X_i = 1] \\
&= \sum_{X : X_i = 1} \Pr[X = (X_1, \ldots, X_{i-1}, 1, X_{i+1}, \ldots, X_n) \mid X_i = 1] \Pr[\hat{X}_i = 1 \mid X] \\
&\leq \sum_{X : X_i = 1} \Pr[X = (X_1, \ldots, X_{i-1}, 0, X_{i+1}, \ldots, X_n) \mid X_i = 0] \Pr[\hat{X}_i = 1 \mid X] \\
&\leq \sum_{X' : X'_i = 0} \Pr[X' = (X'_1, \ldots, 0, \ldots, X'_n)](e^\varepsilon \Pr[\hat{X}_i = 1 \mid X'] + \delta) \\
&= e^\varepsilon \Pr[\hat{X}_i = 1 \mid X_i = 0] + \delta.
\end{aligned}
$$

(c) Similar from part (b), we have

$$\Pr[\hat{X}_i = 0 \mid X_i = 0] \leq e^\varepsilon (1 - \Pr[\hat{X}_i = 1 \mid X_i = 1]) + \delta.$$

Thus,

$$
\begin{aligned}
\mathbb{E}[Z_i] &= \Pr[\hat{X}_i = 1 \mid X_i = 1]p + \Pr[\hat{X}_i = 0 \mid X_i = 0](1 - p) \\
&\leq \Pr[\hat{X}_i = 1 \mid X_i = 1]p + (1 - p)(e^\varepsilon - e^\varepsilon \Pr[\hat{X}_i = 1 \mid X_i = 1]) + (1 - p)\delta \\
&= (1 - p)e^\varepsilon + (p - e^\varepsilon(1 - p)) \Pr[\hat{X}_i = 1 \mid X_i = 1] + (1 - p)\delta \\
&\leq (1 - p)e^\varepsilon + \delta.
\end{aligned}
$$

Then, the average is clearly bounded by the same:

$$\mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} Z_i\right] = \frac{1}{n}\sum_{i=1}^{n} \mathbb{E}[Z_i] \leq (1 - p)e^\varepsilon + \delta.$$

7. Suppose you are given a private dataset $X \in \mathcal{X}^n$, where $\mathcal{X} = \{1, \ldots, d\}$. That is, the dataset X consists of n integers $x_1, \ldots, x_n$ between 1 and d. We want to generate a sample $Y \in \{1, \ldots, d\}$ that satisfies $\min_{i=1}^n x_i \leq Y \leq \max_{i=1}^n x_i$ with high probability, but of course with ε-DP guarantee.

   (a) Define a utility function $q : \mathcal{X}^n \times \mathcal{X} \to \mathbb{Z}$ as:

   $$q(X, y) = \min\{|\{i : x_i \leq y\}|,\ |\{i : x_i \geq y\}|\}$$

   where $|\{\cdot\}|$ denotes the size of the set. Let $\mathrm{OPT}(X) = \max_{y \in \mathcal{X}} q(X, y)$, then show that

   $$\mathrm{OPT}(X) \geq \frac{n - 1}{2}.$$

   (b) We use the exponential mechanism with range $\mathcal{X}$ with utility function q. Find the sensitivity Δ.

(c) Describe an exponential mechanism that satisfies ε-differential privacy, and show that

$$\Pr\left[q(X,Y) \geq \mathrm{OPT}(X) - \frac{2}{\varepsilon}\log\left(\frac{d}{\beta}\right)\right] \geq 1-\beta.$$

You can use the utility guarantee result from the lecture note.

(d) Specify C_1 and C_2 that the above exponential mechanism satisfies

$$\Pr\left[\min_{i=1}^{n} x_i \leq y \leq \max_{i=1}^{n} x_i\right] \geq 1-\beta$$

for

$$n \geq \frac{C_1}{\varepsilon}\ln\left(\frac{d}{\beta}\right) + C_2.$$

Here, C_1 and C_2 should be constants that are independent of n, d, β, and ε. (Hint) Consider the event $q(X,y) \geq 1$.

Solution.

(a) Since we can choose y to be the $\lfloor n/2 \rfloor$-th element of the sorted list, we have

$$\mathrm{OPT}(X) \geq \left\lfloor \frac{n}{2} \right\rfloor \geq \frac{n-1}{2}.$$

(b) We now compute the sensitivity Δ of the utility function q. Let X and X' be neighboring datasets differing in one element. Then for any $y \in \mathcal{X}$,

$$||\{i : x_i \leq y\}| - |\{i : x_i' \leq y\}|| \leq 1,$$
$$||\{i : x_i \geq y\}| - |\{i : x_i' \geq y\}|| \leq 1.$$

Thus,

$$|q(X,y) - q(X',y)| \leq 1,$$

so $\Delta = 1$.

(c) Check the lecture note. We have $\Delta = 1$, $t = \log(1/\beta)$.

(d) Observe that y lies between $\min_i x_i$ and $\max_i x_i$ if and only if $q(X,y) \geq 1$. From part (c), with probability at least $1-\beta$, the output y satisfies:

$$q(X,y) \geq \mathrm{OPT}(X) - \frac{2}{\varepsilon}\ln\left(\frac{|\mathcal{X}|}{\beta}\right).$$

As noted earlier, $\mathrm{OPT}(X) \geq \frac{n-1}{2}$, so if

$$\frac{n-1}{2} - \frac{2}{\varepsilon}\ln\left(\frac{d}{\beta}\right) \geq 1,$$

then $q(X, y) \geq 1$ and therefore y lies between $\min_i x_i$ and $\max_i x_i$.
Solving the inequality,

$$\frac{n-1}{2} \geq \frac{2}{\varepsilon} \ln\left(\frac{d}{\beta}\right) + 1 \quad \Rightarrow \quad n \geq \frac{4}{\varepsilon} \ln\left(\frac{d}{\beta}\right) + 3.$$

Hence, the desired guarantee holds if we set constants $C_1 = 4$ and $C_2 = 3$:

$$n \geq \frac{4}{\varepsilon} \ln\left(\frac{d}{\beta}\right) + 3 \quad \Rightarrow \quad \Pr\left[\min_i x_i \leq y \leq \max_i x_i\right] \geq 1 - \beta.$$